National Tsing Hua University
Department of Electrical Engineering
EE3662 Digital Signal Processing Laboratory, Fall 2025

Lab 5 Pixel Array Manipulation

Assigned on Oct 13, 2025
Due by Oct 19, 2021

Overview

The goal of this homework is to help you get familiar with pixel array manipulation in python. First, you will learn how to input/output image file in python, and then you will need to write some functions to manipulate pixel array, i.e., grayscale, image flipping, image rotation, and image resize in this lab.

Tutorials

1. pip install numpy/imageio
2. Modify the .py file only. (**Do not** need to change main.py or common.py)
3. Run main.py to check the result.
4. Please follow the steps in the source code comments, image processing packages are not allowed to use.
5. Enjoy your homework!

Details

1. Grayscale (10%)

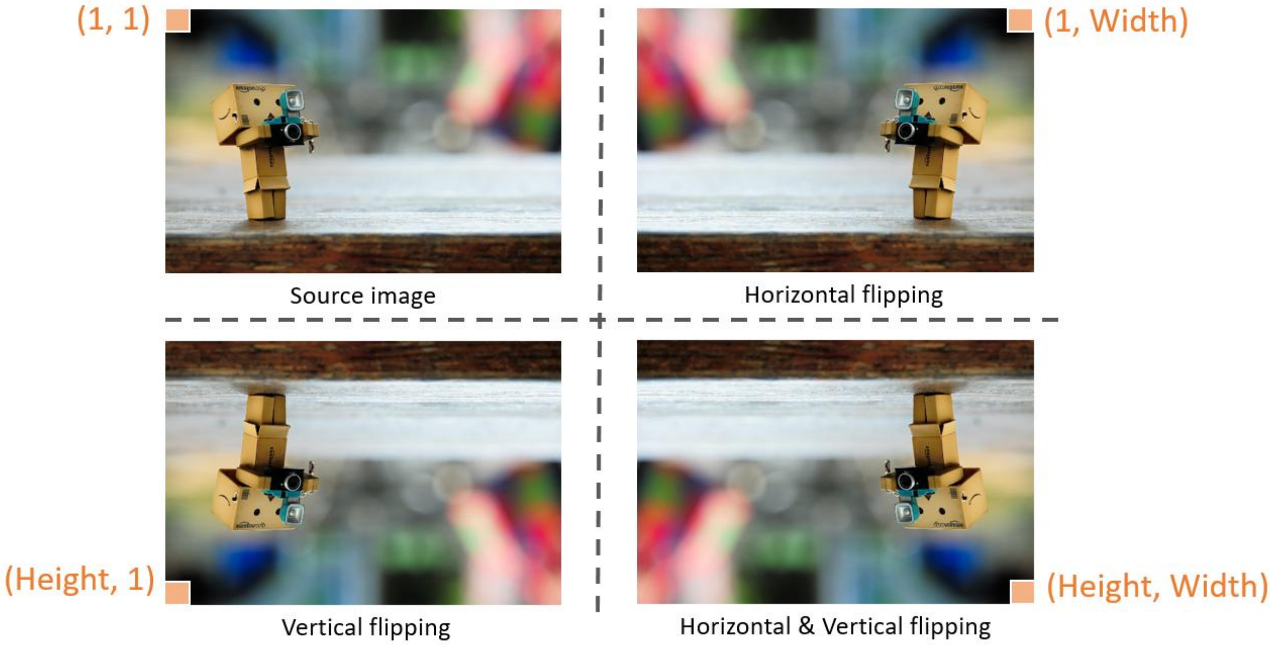
$$Y = [0.299 \quad 0.587 \quad 0.114] * \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

Source image

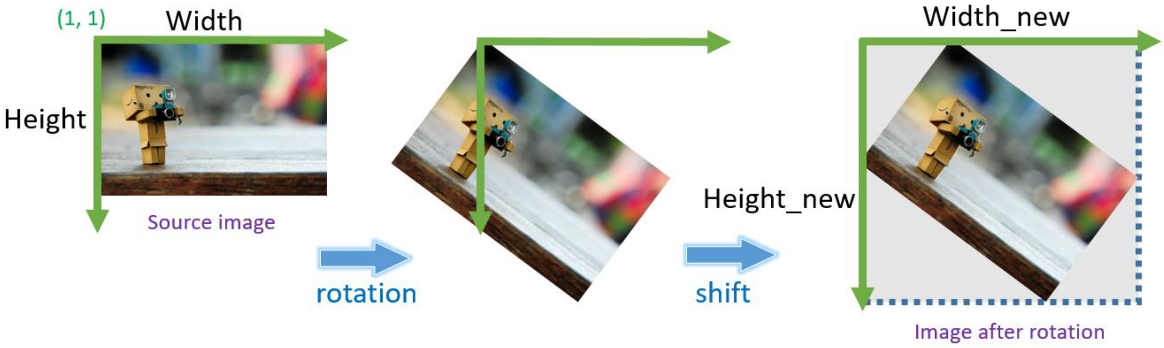
Grayscale

2. Image flipping (10%)

Implement three kinds of flipping.



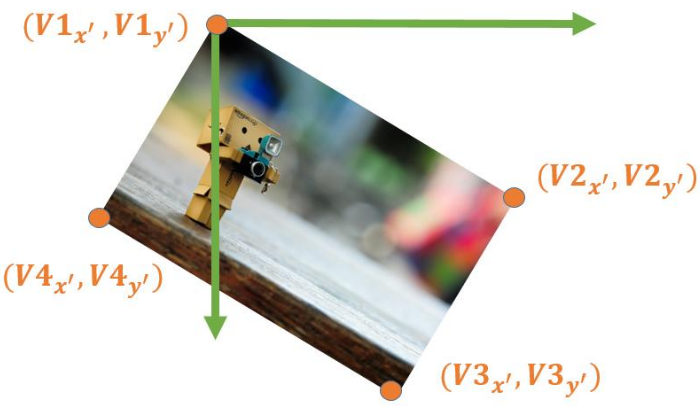
3. Image rotation (15%)



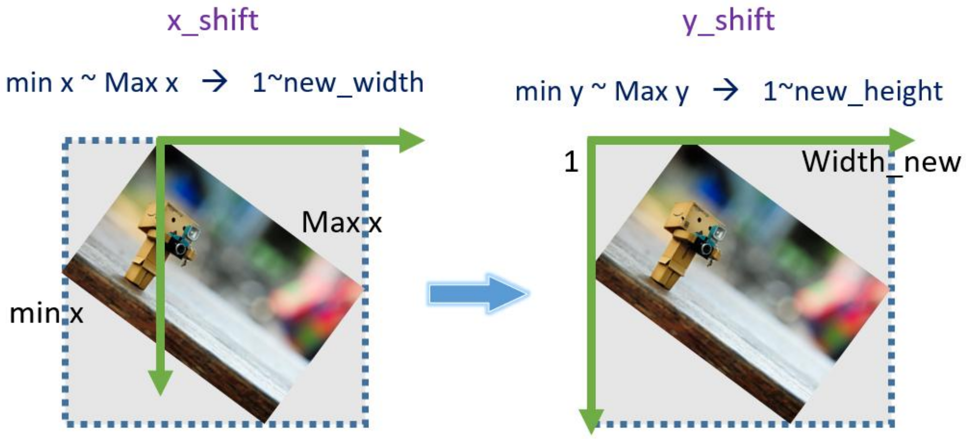
$$Rotation\ Matrix: \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

(1) Record vertices.

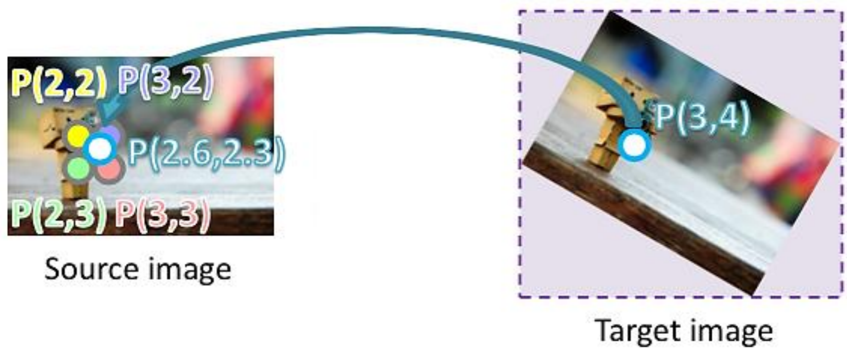
(2) Use rotation matrix to get rotated vertices.



(3) Get new height and width to create new image with zero matrix.

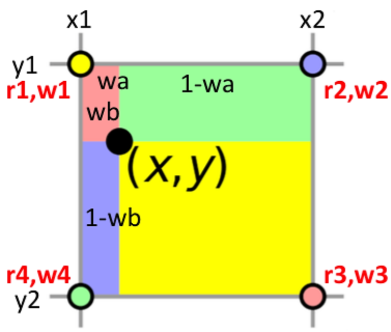


(4) Backward warping.



(5) Bilinear interpolation.

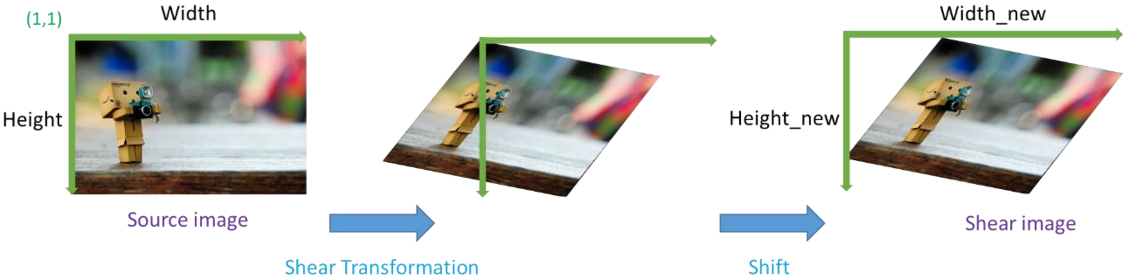
$$
\begin{aligned}
\mathrm{wa} &= (\mathrm{x}-\mathrm{x1}) / (\mathrm{x2}-\mathrm{x1}) \\
\mathrm{wb} &= (\mathrm{y}-\mathrm{y1}) / (\mathrm{y2}-\mathrm{y1}) \\
\mathrm{w1} &= (1-\mathrm{wa}) * (1-\mathrm{wb}) \\
\mathrm{w2} &= \mathrm{wa} * (1-\mathrm{wb}) \\
w3 &= wa * wb \\
\mathrm{w4} &= (1-\mathrm{wa}) * \mathrm{wb}
\end{aligned}
$$



$$
\begin{aligned}
\mathrm{r(x,y)} &= \mathrm{r1} * \mathrm{w1} + \mathrm{r2} * \mathrm{w2} + \mathrm{r3} * \mathrm{w3} + \mathrm{r4} * \mathrm{w4} \\
\mathrm{g(x,y)} &= \mathrm{g1} * \mathrm{w1} + \mathrm{g2} * \mathrm{w2} + \mathrm{g3} * \mathrm{w3} + \mathrm{g4} * \mathrm{w4} \\
\mathrm{b(x,y)} &= \mathrm{b1} * \mathrm{w1} + \mathrm{b2} * \mathrm{w2} + \mathrm{b3} * \mathrm{w3} + \mathrm{b4} * \mathrm{w4}
\end{aligned}
$$

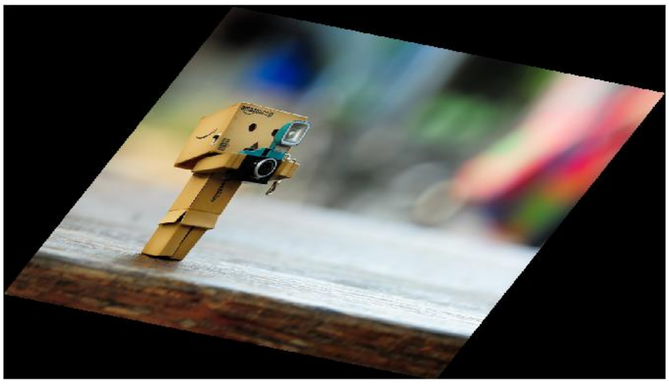
(6) Result image.

4. Shear Transformation (15%)



$$Shear\ Transformation: \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & shear\_x \\ shear\_y & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

(1) Record vertices.
(2) Use transformation matrix to get new vertices.
(3) Get new height and width to create new image with zero matrix.
(4) Backward warping.
(5) Bilinear interpolation.
(6) Result image.

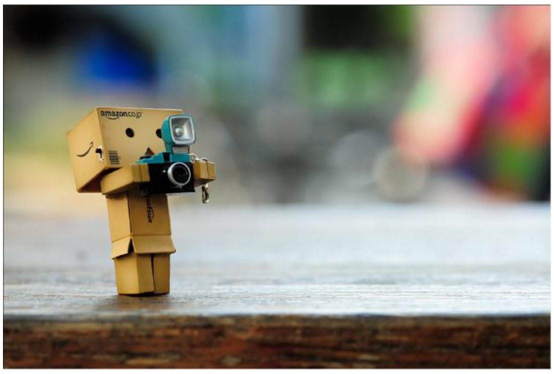


5. Image scaling (10%)

Use the similar interpolation scheme as image rotation to conduct image resize function.

Input image

0.6X



1.5X

Report

1. (7%) Show the results in each part with two images. One is given by TA and the other is your own image.

2. (10%) There are two ways for image warping. One is forward warping and the other is backward warping. What are the differences between them? Try to describe the differences subjectively and objectively.

3. (10%) Try to use the forward warping method mentioned above to implement image rotation. Specify the flow of your implementation and show your results. Please include your code in “LAB5/code/”.

4. (10%) Try to implement at least one other method for grey scale, image flipping, image rotation, or image resize. Specify your method and show the results. Compare the differences between the original method in this lab and your proposed method. Please include your code in “LAB5/code/”.

5. (3%) Conclusion.

Deliverable and file organization

Directory	Filename	Description
LAB5/code/	*.py	All python codes
LAB5/data/	*.png / *.jpg	Your own source image
LAB5/report/	report.pdf	Your report
LAB5/results/	*.png / *.jpg	Your results

Please organize your files according to the above table and compress it as LAB5_<student_id>.zip in ZIP format.